

LEARNING MADE EASY

Snowflake Special Edition

Generative AI and LLMs

for
dummies[®]
A Wiley Brand



Create new business
insights with gen AI

Laying the gen AI base with
a cloud data platform

Boost productivity
with LLMs

Brought to
you by



David Baum

About Snowflake

Snowflake enables every organization to mobilize their data with Snowflake's Data Cloud. Customers use the Data Cloud to unite siloed data, discover and securely share data, power data applications, and execute diverse AI/ML and analytic workloads. Wherever data or users live, Snowflake delivers a single data experience that spans multiple clouds and geographies. Thousands of customers across many industries, including 647 of the 2023 Forbes Global 2000 (G2K) as of October 31, 2023, use Snowflake Data Cloud to power their businesses. Learn more at snowflake.com.



Generative AI and LLMs

Snowflake Special Edition

by David Baum

**for
dummies®**
A Wiley Brand

Generative AI and LLMs For Dummies®, Snowflake Special Edition

Published by
John Wiley & Sons, Inc.
111 River St.
Hoboken, NJ 07030-5774
www.wiley.com

Copyright © 2024 by John Wiley & Sons, Inc., Hoboken, New Jersey

No part of this publication may be reproduced, stored in a retrieval system or transmitted in any form or by any means, electronic, mechanical, photocopying, recording, scanning or otherwise, except as permitted under Sections 107 or 108 of the 1976 United States Copyright Act, without the prior written permission of the Publisher. Requests to the Publisher for permission should be addressed to the Permissions Department, John Wiley & Sons, Inc., 111 River Street, Hoboken, NJ 07030, (201) 748-6011, fax (201) 748-6008, or online at <http://www.wiley.com/go/permissions>.

Trademarks: Wiley, For Dummies, the Dummies Man logo, The Dummies Way, Dummies.com, Making Everything Easier, and related trade dress are trademarks or registered trademarks of John Wiley & Sons, Inc. and/or its affiliates in the United States and other countries, and may not be used without written permission. Snowflake and the Snowflake logo are trademarks or registered trademarks of Snowflake Inc. All other trademarks are the property of their respective owners. John Wiley & Sons, Inc., is not associated with any product or vendor mentioned in this book.

LIMIT OF LIABILITY/DISCLAIMER OF WARRANTY: WHILE THE PUBLISHER AND AUTHORS HAVE USED THEIR BEST EFFORTS IN PREPARING THIS WORK, THEY MAKE NO REPRESENTATIONS OR WARRANTIES WITH RESPECT TO THE ACCURACY OR COMPLETENESS OF THE CONTENTS OF THIS WORK AND SPECIFICALLY DISCLAIM ALL WARRANTIES, INCLUDING WITHOUT LIMITATION ANY IMPLIED WARRANTIES OF MERCHANTABILITY OR FITNESS FOR A PARTICULAR PURPOSE. NO WARRANTY MAY BE CREATED OR EXTENDED BY SALES REPRESENTATIVES, WRITTEN SALES MATERIALS OR PROMOTIONAL STATEMENTS FOR THIS WORK. THE FACT THAT AN ORGANIZATION, WEBSITE, OR PRODUCT IS REFERRED TO IN THIS WORK AS A CITATION AND/OR POTENTIAL SOURCE OF FURTHER INFORMATION DOES NOT MEAN THAT THE PUBLISHER AND AUTHORS ENDORSE THE INFORMATION OR SERVICES THE ORGANIZATION, WEBSITE, OR PRODUCT MAY PROVIDE OR RECOMMENDATIONS IT MAY MAKE. THIS WORK IS SOLD WITH THE UNDERSTANDING THAT THE PUBLISHER IS NOT ENGAGED IN RENDERING PROFESSIONAL SERVICES. THE ADVICE AND STRATEGIES CONTAINED HEREIN MAY NOT BE SUITABLE FOR YOUR SITUATION. YOU SHOULD CONSULT WITH A SPECIALIST WHERE APPROPRIATE. FURTHER, READERS SHOULD BE AWARE THAT WEBSITES LISTED IN THIS WORK MAY HAVE CHANGED OR DISAPPEARED BETWEEN WHEN THIS WORK WAS WRITTEN AND WHEN IT IS READ. NEITHER THE PUBLISHER NOR AUTHORS SHALL BE LIABLE FOR ANY LOSS OF PROFIT OR ANY OTHER COMMERCIAL DAMAGES, INCLUDING BUT NOT LIMITED TO SPECIAL, INCIDENTAL, CONSEQUENTIAL, OR OTHER DAMAGES.

For general information on our other products and services, or how to create a custom *For Dummies* book for your business or organization, please contact our Business Development Department in the U.S. at 877-409-4177, contact info@dummies.biz, or visit www.wiley.com/go/custompub. For information about licensing the *For Dummies* brand for products or services, contact BrandedRights&Licenses@Wiley.com.

ISBN 978-1-394-23842-2 (pbk); ISBN 978-1-394-23843-9 (ebk)

Publisher's Acknowledgments

Some of the people who helped bring this book to market include the following:

Development Editor: Nicole Sholly

Sales Manager: Molly Daugherty

Project Manager: Jennifer Bingham

Content Refinement Specialist:

Acquisitions Editor: Traci Martin

Saikarthick Kumarasamy

Editorial Manager: Rev Mengle

Table of Contents

INTRODUCTION 1

- About This Book 1
- Icons Used in This Book..... 2
- Beyond the Book..... 2

CHAPTER 1: Introducing Gen AI and the Role of Data 3

- The Historical Context of Gen AI 3
 - Introducing LLMs and foundation models 4
 - Transforming the AI landscape 5
 - Accelerating AI functions 5
- The Role of Data in AI Projects 6
- Explaining the Importance of Generative AI to the Enterprise..... 7
 - Pretrained models 8
 - Security versus ease of use 9
- Managing Gen AI Projects with a Cloud Data Platform 10

CHAPTER 2: Understanding Large Language Models 11

- Categorizing LLMs 11
 - Defining general-purpose LLMs..... 12
 - Using task-specific and domain-specific LLMs 14
- Reviewing the Technology Behind LLMs 14
 - Introducing key terms and concepts..... 15
 - Explaining the importance of vector embeddings..... 16
 - Identifying developer tools and frameworks 17
 - Enforcing data governance and security 17
 - Extending governance for all data types..... 18

CHAPTER 3: LLM App Project Lifecycle 19

- Defining the Use Case and Scope 19
 - Selecting the right LLM..... 20
 - Comparing small and large language models..... 21
- Adapting LLMs to Your Use Case..... 22
 - Engineering prompts..... 22
 - Learning from context..... 23
 - Augmenting text retrieval 23
 - Fine-tuning language models 24
 - Reinforcement learning 25
 - Using a vector database..... 25